

The Quantum Routing Brain

Orchestrating Quantum Microservices
in Tactical Defense Systems

Challenge 2 | Vanderbilt Dry Dock 2026 | Quantum & Frontier Tech

Challenge Sponsor

Xtremis, Inc. (d/b/a Xtremis AI)

Washington, DC | Devil's Den, AR | Nashville, TN

Point of Contact

B. Thomas Edman, Strategy & Technology Lead

Available remotely during Build Week for office hours



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THE PROBLEM

Every modern military system—counter-drone kill chains, logistics networks, sensor fusion pipelines, coalition communications—is a stream of compute tasks running across edge devices, tactical clouds, and rear-area data centers. Within the next 5–10 years, some of those tasks will be candidates for quantum acceleration: combinatorial optimization, certain classification problems, random number generation for cryptographic keying. But quantum resources will be scarce, expensive, remote, and constrained in ways classical compute is not. Commercial QPUs today operate at most at Impact Level 5 (Controlled Unclassified Information). Anything SECRET or above cannot leave the tactical enclave. Tactical networks are Disconnected, Degraded, Intermittent, and Limited (DDIL): a round trip to a CONUS quantum cloud takes 500+ ms over GEO SATCOM, and the link may drop mid-computation. Nobody has built the middleware that makes quantum a first-class citizen in this environment.

THE THESIS

Backend providers—IBM, IonQ, AWS, Quantinuum—will commoditize each other. The orchestration intelligence does not commoditize. The company that owns the policy-aware, classification-aware, DDIL-tolerant routing layer becomes the integration fabric for quantum-defense. Build the control plane now; swap in better quantum backends as the hardware matures.

Honesty is rewarded. In 2026, the correct default for nearly every operational task is “run classical.” The broker’s near-term value is not deciding when quantum wins. It is being the layer that lets a defense customer adopt quantum incrementally without re-architecting their tactical stack each time the hardware improves.

IL5 max commercial QPU accreditation | **500+ ms** GEO SATCOM round trip | **~1 of 47** defense tasks
quantum-ready today | **0** auto-dispatchers in production

YOUR MISSION

Build a quantum job broker for tactical defense workflows. Pick a mission thread from the four below. Map its computational steps. Identify where quantum sub-problems surface. Build the middleware that routes tasks intelligently between classical and quantum backends, respecting classification, latency, and network constraints.

The deliverable is not a quantum algorithm. It is the *orchestration layer around* the quantum algorithm—the software that decides what runs where, enforces policy, handles degradation, and logs everything for audit.

PICK YOUR MISSION THREAD

Choose one of the following scenarios as your grounding use case. Each describes an operational workflow with computational steps where quantum sub-problems naturally surface. Your venture is the orchestration layer, not the mission application itself. The thread grounds your demo in a real operational context.

Thread 1: Counter-UAS

Scenario. A forward-deployed sensor network detects an RF emission from an unidentified small drone. The system must classify the drone type from its RF signature, geolocate it via multi-sensor fusion (TDOA/AOA from distributed receivers), correlate the track against known platforms, assign an effector (jammer, kinetic, or directed energy), and execute—all within seconds.

Computational steps: FFT/spectrogram on raw I/Q, CFAR detection, specific emitter identification, modulation classification, TDOA/AOA geolocation (matrix inversion), multi-hypothesis track correlation (combinatorial assignment), weapon-target assignment (optimization).

Where quantum surfaces: Emitter classification (quantum kernel SVM on I/Q features), track-to-effector assignment (QUBO formulation of the weapon-target assignment problem), geolocation (HHL-class linear system, fault-tolerant only).

Thread 2: Satellite Constellation Tasking & Orbital Deconfliction

Scenario. A space operations center manages 40+ imaging and SIGINT satellites. Each collection cycle, hundreds of competing tasking requests arrive from combatant commands, intelligence agencies, and coalition partners—each with priority levels, time windows, off-nadir angle constraints, cloud cover forecasts, and revisit requirements. The scheduler must maximize collection value while respecting orbital mechanics (slew rates, power budgets, downlink windows), deconflicting with 40,000+ tracked objects, and re-planning in near-real-time when a satellite degrades or a priority-one request preempts the queue.

Computational steps: Orbit propagation (SGP4/SDP4), access window calculation, multi-objective scheduling with constraints (time, power, thermal, storage), collision avoidance maneuver planning, downlink scheduling across a limited ground station network, priority arbitration across classification levels.

Where quantum surfaces: Constellation tasking as a multi-dimensional knapsack/scheduling QUBO (hundreds of binary variables), collision avoidance trajectory optimization, ground station downlink scheduling (graph coloring on time-frequency-antenna resources).

Thread 3: Cryptographic Key Distribution & PQC Migration for Coalition Networks

Scenario. A multinational task force operates across five nations with different cryptographic policies, key management infrastructures, and classification regimes. Each nation's tactical network must exchange targeting data, logistics updates, and ISR products at machine speed, but the cryptographic handshakes protecting these exchanges are approaching quantum vulnerability. The system must manage the transition from classical (RSA/ECDH) to post-quantum (ML-KEM, ML-DSA) key encapsulation and signing across heterogeneous national enclaves while maintaining backward compatibility with legacy systems that cannot be upgraded in the field.

Computational steps: Key generation and distribution across multi-national enclaves, certificate validation chain construction across national PKIs, algorithm negotiation (hybrid classical+PQC during transition), PQC performance profiling on constrained tactical hardware, QRNG integration for entropy at the edge, cryptographic agility layer that hot-swaps algorithms without breaking sessions.

Where quantum surfaces: QRNG for keying material (the single operationally deployed quantum technology today, NIST SP 800-90B validated April 2025), PQC algorithm selection routing ("which PQC primitive runs fastest on this specific edge device"), key distribution network optimization (graph optimization on the multi-national trust topology as QUBO).

Thread 4: Contested Logistics & Resupply Optimization

Scenario. A theater sustainment command must plan daily resupply convoys across a road network where bridges are interdicted, routes are contested by adversary fires, fuel is constrained, and vehicle availability changes hourly. The optimization must account for time windows at delivery points, vehicle capacity, threat exposure (minimize time in adversary sensor/fire envelopes), and priority of supply classes. Re-planning must happen in near-real-time as conditions change.

Computational steps: Vehicle routing with time windows and capacity constraints (VRP, NP-hard), threat-exposure cost modeling on road segments, fuel consumption optimization, stochastic re-planning as routes are interdicted, convoy scheduling across multiple echelons.

Where quantum surfaces: VRP as QUBO (the most-studied quantum optimization problem after MaxCut, with published demonstrations at 5–13 nodes), threat-path optimization (shortest path with adversarial edge weights), demand forecasting (quantum kernel regression on supply consumption patterns).

WHAT YOU BUILD

Your core deliverable is a working quantum job broker with three components:

1. The Router

A service that ingests a task description from your chosen mission thread, evaluates it against a decision rubric (Is this task quantum-eligible? Does the classification level permit offload? Does the latency budget allow a round trip? Is a quantum backend available?), and routes it to the appropriate backend. The routing decision must be explainable and logged.

2. The Backends

At minimum two: a classical solver (brute force, Gurobi, OR-Tools, or scikit-learn depending on the problem) and a quantum solver (Qiskit Aer simulator, with optional stretch to IBM Open Plan hardware or AWS Braket). Both wrapped as containerized services behind a common API so the router treats them interchangeably.

3. The Data Guard (Stretch Goal)

A middleware component that inspects outbound quantum job requests for classification-sensitive content, strips or redacts it before transmission to a commercial (IL5-max) QPU, and reattaches context on result return. Even a stub that demonstrates the pattern—flag fields by classification, redact before send, reattach after receive—is valuable. This is the hardest and most novel component.

TACTICAL CONSTRAINTS REFERENCE

Your routing engine operates in a constrained environment. The table below gives realistic parameters.

Link Type	Round-Trip Latency	Bandwidth
GEO MILSATCOM (WGS, AEHF)	500–600 ms	1–10 Mbps tactical
LEO (Starlink/Starshield)	20–50 ms	100–150 Mbps
MUOS (UHF SATCOM)	GEO-class	Up to 384 kbps/user
Link 16 / TADIL-J	LOS, slot-bounded	31.6–115.2 kbps
Tactical mesh (Silvus/PW)	<45 ms node-to-node	Up to ~100 Mbps
Tactical 5G (fixed base)	<15 ms	1.5–3.9 Gbps demonstrated

Classification ceiling. Commercial quantum services (AWS Braket, Azure Quantum, IBM Quantum) operate at most at Impact Level 5 (CUI). No commercial QPU is currently authorized at IL6 (SECRET). This is the structural reason your routing engine must gate offload decisions by classification level.

WHAT SUCCESS LOOKS LIKE

Criterion	Summary	Weight
Technical Substance	Does the work demonstrate meaningful engagement with the orchestration problem? A working broker on a free-tier quantum SDK with runtime extrapolation, or an entirely classical build with genuine quantitative analysis of where and when quantum plugs in, are both valid. Overclaiming quantum advantage will hurt you. Honest framing will help.	35%
Venture Strategy	Not a pitch deck exercise. Show us the next 12–24 months: prototype vs. MVP vs. product gates, capital needs at each stage, non-dilutive financing pathways (SBIR, STTR, NSF, DARPA), first design partners, expert hiring strategy, and how you maintain equity while growing.	25%
Presentation & Narrative	Can you hold a room and answer hard questions without retreating to buzzwords? We will ask you technical questions about quantum computing and tactical systems to test whether you understand the problem or outsourced the thinking.	15%
Engineering Quality	Six days is not long. We expect clean, auditable practices: readable code, structured repository, containerization where appropriate, and evidence that AI-generated code was reviewed and understood.	15%
Team Execution	Did the team divide work intelligently? Can each person speak with authority on their piece? We score teams and watch individuals. Tell us what you led, not your title.	10%

Full rubric with detailed expectations for each criterion is available in the shared Annex: **Evaluation Rubric**.

RESOURCES PROVIDED

The following materials are available in the challenge repository:

- **Annex A: Task Profile CSV.** ~25 canonical defense compute tasks with classical baselines, quantum candidacy flags, classification levels, and latency budgets.
- **Annex B: Decision Rubric.** "When is quantum worth the round trip?" Covers problem class, instance size, classification gating, backend availability, and the honest 2026 default.
- **Annex C: Free Quantum Resource Guide.** IBM Open Plan, AWS Braket, Qiskit Aer, PennyLane, D-Wave Leap. What is free, realistic limits, and recommended hackathon stack.
- **Annex D: DDIL & Classification Reference.** Tactical link budgets, IL2/IL4/IL5/IL6 definitions, commercial QPU accreditation ceiling, CoT message format.
- **Annex E: DevSecOps Reference Pointers.** DoD DevSecOps Reference Design, Platform One Big Bang (GitHub), K3s, container signing toolchain.
- **Annex F: Evaluation Rubric.** Full rubric with detailed expectations for all five scoring criteria. Shared across both Xtremis challenges.
- **Starter Repository.** Docker Compose scaffold (router + classical backend + quantum backend), the task CSV, and a reference MaxCut QAOA script as the pluggable quantum kernel.



Make the Invisible Actionable

Persistent spectrum awareness | AI-driven orchestration | Operationalized at scale

Questions during Build Week:

B. Thomas Edman | btedman@xtremis.ai

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Prepared for Vanderbilt Dry Dock 2026 Build Week.